



Digital Assistant for Classroom Management

Project Documentation

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Department: Artificial Intelligence and Data Science

Institution: Bannari Amman Institute of Technology

Academic Year: 2025 - 2026

Generated from the current final implementation of the web application.

Digital Assistant for Classroom Management

Title Page

- Project Title: Digital Assistant for Classroom Management
- Developer Name: Poornesh S
- Institution: Bannari Amman Institute of Technology, Sathyamangalam
- Department: Artificial Intelligence and Data Science
- Academic Year: 2025 - 2026

Abstract

Digital Assistant for Classroom Management is a web-based academic monitoring system developed for faculty members. The application consolidates student records, attendance, semester performance, AI-assisted risk analysis, leaderboard ranking, department analytics, activity logging, and export workflows into one responsive platform. The final version focuses on day-to-day usability, clear data presentation, and safer handling of invalid input and service failures.

Introduction

Traditional classroom monitoring often depends on spreadsheets, disconnected records, and manual review of attendance and marks. This slows down academic decision-making and makes it difficult for faculty to identify weak students early. The current system addresses that problem through a centralized web interface where student performance, department trends, and risk signals can be reviewed in a consistent and structured way.

Objectives

- Build a secure faculty-facing classroom management web application.
- Maintain student records with validation and structured academic data entry.
- Provide dashboard-level analytics for faster academic monitoring.
- Highlight weak students using attendance and performance indicators.
- Offer detailed student profile pages with semester-wise charts and AI insights.
- Support leaderboard and department-level review workflows.
- Maintain activity logs for important classroom operations.
- Enable report export for student and department analysis.

System Architecture

The overall application flow follows this structure:

- User

- Web Interface
- Flask Backend
- Data Storage Layer
- Analysis and Reporting Layer
- Dashboard and Profile Views

In practical terms, the system works as:

- User requests are sent through HTML, CSS, and JavaScript based pages.
- Flask blueprints process login, page rendering, and API calls.
- Student and course data are read from CSV files.
- AI log history is stored in SQLite.
- Feedback data is stored in JSON format.
- Service modules prepare analytics, weak student indicators, rankings, and chart data.
- The processed results are rendered in dashboard, student profile, leaderboard, and activity log pages.

Technology Stack

- Frontend: HTML, CSS, JavaScript, Jinja2
- Backend: Python, Flask, Flask-Compress
- Data Processing: Pandas, NumPy, scikit-learn
- Visualization: Matplotlib
- Reporting: ReportLab
- Storage: CSV, JSON, SQLite
- Deployment Target: Render using Gunicorn

Implementation Details

Login System

The application uses a faculty login flow with session-based authentication. Protected pages redirect unauthenticated users to the login page. Logout is confirmed on the client side and the session is cleared safely.

Dashboard Analytics

The dashboard presents high-level academic visibility with:

- total student count
- weak student count

- average attendance
- risk alerts
- department cards showing total and weak student counts
- a performance table with grades and predictions

The dashboard also includes animated counters, loading states, and responsive layout behavior.

Student Management

The student management module supports:

- add student
- edit student
- delete student
- bulk delete
- search by name or register number
- department filtering
- sort by marks, attendance, and weak-student priority
- CSV export
- profile image upload

The student form uses validation for semester, attendance, unique register number, and subject marks.

Student Profile Analysis

Each student profile provides:

- personal and department details
- current semester
- performance index
- department rank
- improvement summary
- semester-wise marks table
- attendance analysis
- performance trend chart
- semester contribution chart
- AI insight area with live analysis button
- student PDF export

Charts are generated dynamically using Matplotlib and stored inside the static charts directory.

Leaderboard

The leaderboard module ranks students by current semester performance. It includes:

- overall topper card
- department filtering
- top five and lowest five views
- direct links from leaderboard rows to student profiles

Weak Student Detection

Weak student identification is based on low semester totals, attendance thresholds, and risk conditions derived from processed academic data. These indicators are visible on the dashboard, department views, and student profiles.

Activity Logs

The activity log page records key faculty actions such as:

- login and logout
- student add, edit, delete, and bulk delete
- feedback submission
- AI analysis activity
- CSV export
- PDF export

This provides an audit-friendly history of important classroom workflows.

AI Analysis and AI Logs

The student profile page can trigger AI analysis through the API layer. The AI service returns:

- risk level
- performance score
- average marks
- attendance

The AI logs page displays stored AI responses from SQLite and supports filtering by student name, row limit selection, and JSON API access. API processing includes timeout protection and safe JSON error responses.

Course and Feedback Management

The final implementation also includes:

- course add, search, edit, and delete workflows
- feedback form with name, email, message, and rating validation
- JSON-based feedback storage

UI Improvements

The final website includes several user interface refinements:

- dark mode with saved preference in local storage
- responsive navigation and layout behavior for smaller screens
- loading overlays for long-running actions
- fade-in and reveal animations
- auto-dismiss alerts
- confirmation prompts for deletion and logout actions

Database Structure

The application uses a mixed local storage model:

- ``database/students.csv`` stores student academic records, attendance, semester totals, register numbers, departments, and course marks.
- ``database/courses.csv`` stores active course codes and names.
- ``database/app.db`` stores AI analysis logs in the ``ai_logs`` SQLite table.
- ``data/feedback.json`` stores submitted feedback entries.
- ``activity_log.txt`` stores activity history in text form.

This structure works well for the current project scope and local demonstration environment.

Screenshots

The PDF generator automatically embeds screenshots found in ``docs/screenshots/``.

The current documentation includes screenshots for:

- Login page
- Dashboard
- Student Management
- Leaderboard
- AI Logs
- Feedback

- Courses

Advantages of the System

- Centralized academic monitoring in one interface
- Faster identification of weak and at-risk students
- Better faculty visibility into attendance and performance trends
- Export-ready reports for student and department review
- Audit support through activity logging
- Improved usability through responsive layout, dark mode, and loading feedback

Future Enhancements

- Mobile application support
- Parent and student access portal
- Managed cloud database integration
- Role-based access control for faculty and administrators
- Advanced predictive models with richer academic history
- Notification and alert workflows for at-risk students

Conclusion

Digital Assistant for Classroom Management improves classroom administration by combining student record maintenance, academic analytics, AI-assisted insights, activity tracking, and export workflows in a single faculty-oriented web application. The final version is more polished, responsive, and deployment-ready, while still remaining simple enough for academic demonstration and future extension.

Embedded Website Screenshots



**Bannari Amman
Institute of
Technology**
Digital Classroom
Assistant

Dashboard

Student Management

Leaderboard

Activity

AI Logs

Light Mode

Logout

Courses

Feedback

Admin Activity Log

Audited actions for student management workflows.

Timestamp	User	Action	Details
2026-03-12 08:52	admin	Faculty Logged In	Successful login
2026-03-12 08:44	admin	Faculty Logged In	Successful login
2026-03-12 08:41	admin	Faculty Logged In	Successful login
2026-03-12 08:39	admin	Faculty Logged In	Successful login
2026-03-12 08:38	admin	Viewed Student Profile	Poornesh (7376232AD209)
2026-03-12 08:38	admin	Exported PDF For Student	Poornesh
2026-03-12 08:38	admin	Viewed Student Profile	Poornesh (7376232AD209)
2026-03-12 08:37	admin	Faculty Logged In	Successful login
2026-03-12 08:35	admin	Faculty Logged Out	Session ended

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Activities



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Digital Classroom Assistant

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After Login



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Assistant

Dashboard

Student Management

Leaderboard

Activity

AI Logs

Dark Mode

Logout

Courses

Feedback

AI Analysis Logs

Stored responses from the student risk analyzer. Demo data is seeded automatically when no logs exist.

Student Name

Rows

Filter by name

100

Apply Filters

Open JSON API

ID	Student	Risk Level	Performance Score	Timestamp
5	Aarav Kumar	High Risk	9.14	2026-03-12 06:30:49
4	Aarav Kumar	High Risk	9.14	2026-03-12 06:30:49
3	Rahul Nair	High	4.8	2026-03-07 04:26:29
2	Diya Mishra	Medium	6.9	2026-03-07 04:26:29
1	Aarav Kumar	Low	8.7	2026-03-07 04:26:29

Ai Logs

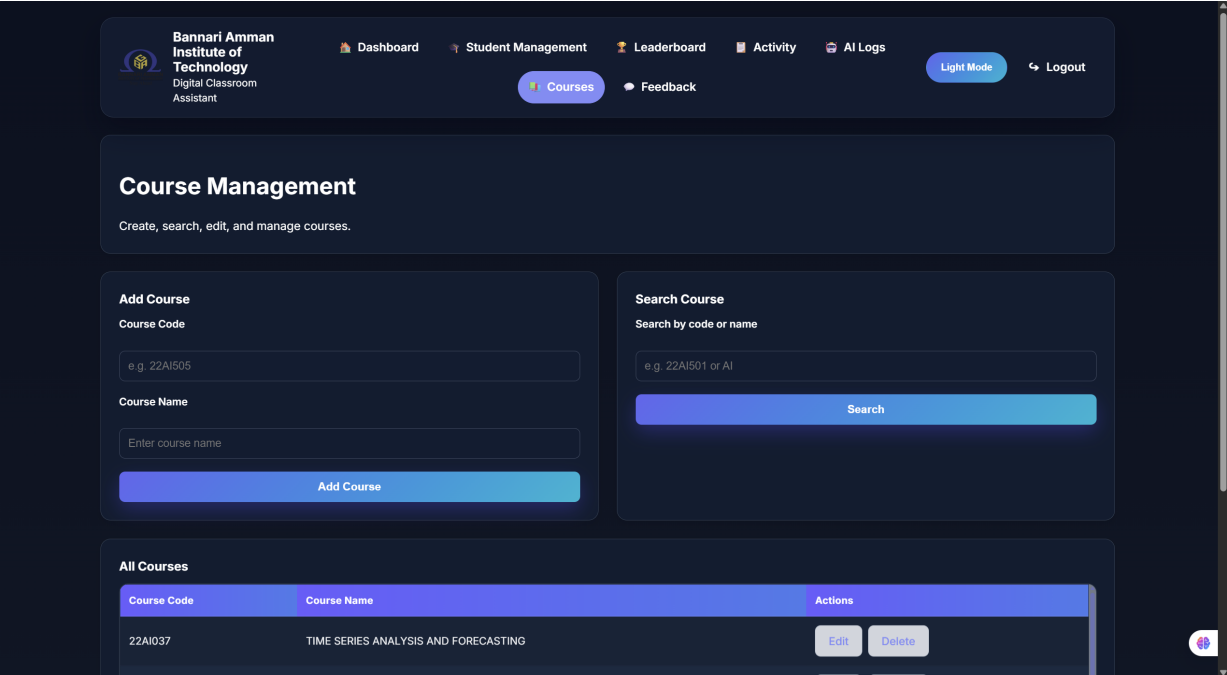


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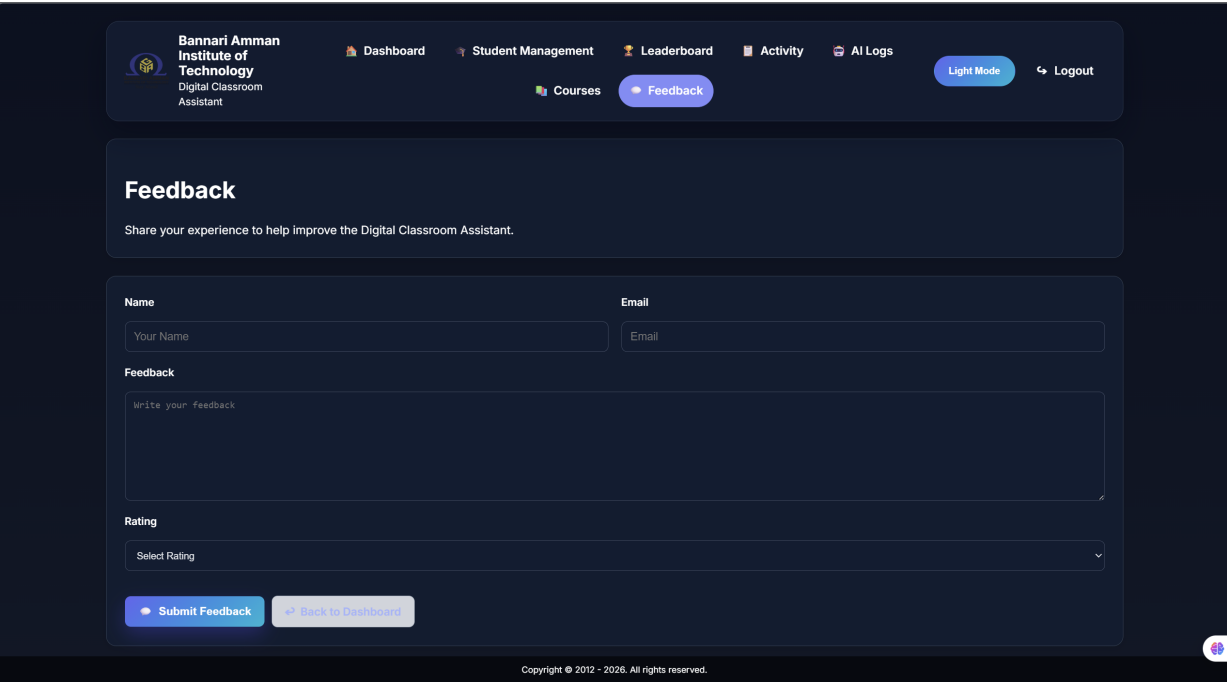
Digital Classroom Assistant

AI Powered Academic Performance Monitoring

Before Login



Courses



Feedback

**Overall Topper**

Aditya Patel

Computer Technology
500 marks

Rank	Name	Department	Sem6 Total
1	Aditya Patel	Computer Technology	500
2	Poornesh	Artificial Intelligence and Data Science	500
3	Reyansh Iyer	Electronics & Instrumentation Engineering	500
4	Rohan Sharma	Computer Science & Engineering	500
5	Siddharth Reddy	Artificial Intelligence and Machine Learning	499

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Leader Board

Empowering Digital Learning

Smart classroom management for modern institutions

AI

CLD

**Faculty Login**
Digital Classroom Assistant

Username

Enter username

Password

Enter password

Login

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Login Page



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Digital Classroom

Assistant

Dashboard

Student Management

Leaderboard

Activity

AI Logs

Courses

Feedback

Light Mode

Logout

Student Management

Manage student records from a full page using the existing classroom dashboard style.

Search Students

Department

Sort By

Search by name or register number

All

Name

Apply

Reset

+ Add Student

Export CSV

Delete Selected

Select	Profile Picture	Student Name	Register Number	Department	Semester	Attendance	Actions
☐		Aadhya Iyer	INFO2026073	Information Science & Engineering	6	84%	View Profile Edit Delete
☐		Aadhya Kumar	CIVI2026013	Civil Engineering	6	98%	View Profile Edit Delete

Student Management